

Spring 2021 Real Analysis II — Homework 2

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1. Let $X := \{x \in C((0, 1]) : \|x\| < \infty\}$ where $\|x\| := \sup_{0 < t \leq 1} |x(t)|$. Show that $\|\cdot\|$ is a norm on X .
2. Let $X = C^1([-1, 1])$ and $\|x\| := (\int_{-1}^1 (|x(t)|^2 + |x'(t)|^2) dx)^{1/2}$ for any $x \in X$. Show that $\|\cdot\|$ is a norm on X , but X is not complete under $\|\cdot\|$. [Hint: consider $x_k(t) = (t^2 + \frac{1}{k^2})^{1/2}$ for $-1 \leq t \leq 1$ and $k \in \mathbb{N}$ and $x(t) = |t|$.]

3. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms defined on $C([0, 1])$ as follows:

$$\|f\|_1 = \left(\int_0^1 |f(x)|^2 dx \right)^{1/2}, \quad \|f\|_2 = \left(\int_0^1 (1+x)|f(x)|^2 dx \right)^{1/2}.$$

Show that these two norms are equivalent in $C([0, 1])$.

4. Let $X := \{f \in C([0, \infty)) : \exists M > 0, \text{ s.t. } |f(x)| \leq M, \forall x \geq 0\}$. For any $a > 0$, define

$$\|f\|_a := \left(\int_0^\infty e^{-ax} |f(x)|^2 dx \right)^{1/2}.$$

Show that $\|\cdot\|_a$ is a norm on X .

5. Consider the same setting as in Problem 4. Suppose $b > a > 0$, show that $\|\cdot\|_b$ and $\|\cdot\|_a$ are not equivalent. [Hint: it suffices to show that there exists $\{f_k\}$ such that $\|f_k\|_a \geq k\|f_k\|_b$ for all k sufficiently large. For example, consider $f_k(x)$ set to $e^{bx/2}$ on $[0, k]$, $-e^{bk/2}(x-k-1)$ on $(k, k+1]$, and 0 on $(k+1, \infty)$.]
6. Let $(X, \|\cdot\|)$ be a B^* space. Show that X is complete if and only if $\sum_{k=1}^\infty x_k$ converges whenever the sequence $\{x_k\}$ satisfies $\sum_{k=1}^\infty \|x_k\| < \infty$. [Hint: use the fact that a Cauchy sequence is convergent if and only if a subsequence is convergent.]
7. Let X be a B^* space and $f : X \rightarrow \mathbb{R}$ be convex, i.e.,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y), \quad \forall \lambda \in [0, 1], x, y \in X.$$

Suppose x^* is a local minimizer of f , i.e., there exists $\delta > 0$ such that $f(x^*) = \min_{x \in B(x^*; \delta)} f(x)$. Show that x^* is a global minimizer of f , i.e., $f(x^*) \leq f(x)$ for all $x \in X$.

8. Let X be a B^* space and M a proper finite dimensional subspace of X . Show that there exists $y \in X$ such that $\|y\| = 1$ and $\|y - x\| \geq 1$ for all $x \in M$.
9. Let $C_0 := \{x = (\xi_1, \xi_2, \dots) : \lim_{k \rightarrow \infty} \xi_k = 0, \xi_k \in \mathbb{R}, \forall k \in \mathbb{N}\}$, i.e., C_0 is the set of real sequences that converge to 0. Define the norm $\|x\| = \sup_{k \in \mathbb{N}} |\xi_k|$. Let $M := \{x = (\xi_1, \xi_2, \dots) \in C_0 : \sum_{k=1}^\infty \frac{\xi_k}{2^k} = 0\}$. Show that M is a closed subspace of C_0 .

10. Consider the same setting as Problem 9. Let $x_0 = (2, 0, 0, \dots)$. Show that

$$\inf_{z \in M} \|x_0 - z\| = 1,$$

but $\|x_0 - z\| > 1$ for all $z \in M$. [Hint: consider $z^{(n)} = (1 - \frac{1}{2^n}, -1, \dots, -1, 0, \dots)$ with the first n terms being nonzero. Remark: this example shows that minimum may not be achieved if M is infinite dimensional.]